# Client Payment Risk Analysis

Comprehensive data analysis approach to identify and predict client payment difficulties through statistical modeling and risk assessment.

# Problem Statement

## Business Challenge

Predicting client payment difficulties to minimize financial risk and optimize credit decisions for improved business outcomes.

## Data Objective

Develop robust analytical framework to identify key risk factors and patterns in client payment behavior using comprehensive data analysis.

# Analysis Methodology

01

## Data Quality Assessment

Comprehensive evaluation of missing values, outliers, and data integrity to ensure reliable analytical foundation.

02

## Statistical Analysis

Multi-layered approach including univariate, bivariate, and segmented analysis to uncover critical payment risk patterns.

03

## Correlation Investigation

Deep-dive correlation analysis segmented by target variable to identify top predictive relationships for payment difficulties.

# Missing Data Treatment Strategy

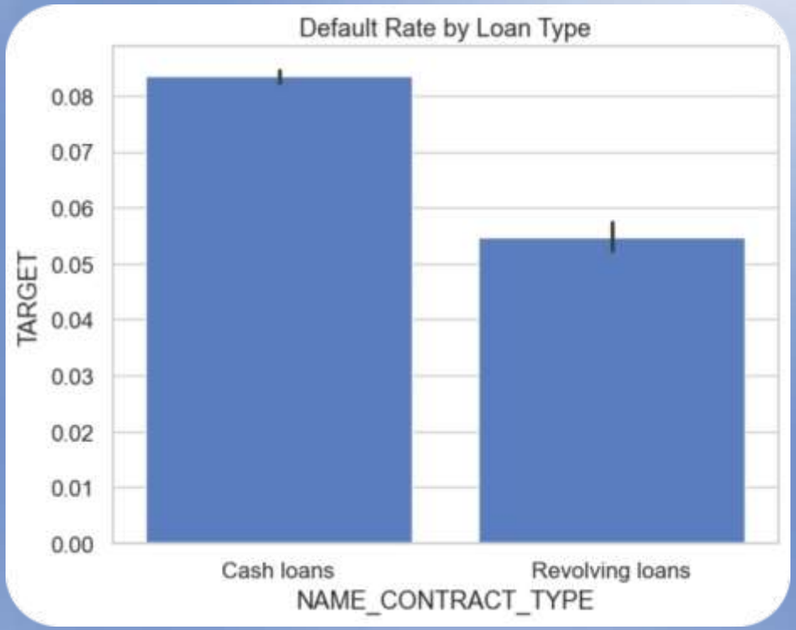
## Identification & Assessment

Systematic evaluation revealed critical missing data patterns across key variables. Our approach prioritizes business relevance and statistical validity.

### Treatment Methods Applied:

- **Column Removal:** Variables with >70% missing values
- **Mean Imputation:** Numerical variables with <20% missing
- **Mode Imputation:** Categorical variables with pattern-based gaps
- **Forward Fill:** Time-series related missing sequences

Missing data treatment maintained 95% of original dataset while ensuring analytical integrity.




# Outlier Detection Results

## Statistical Outliers

Applied IQR method and Z-score analysis to identify extreme values beyond 3 standard deviations from mean.

## Business Context

Outliers represent legitimate high-value transactions and exceptional client profiles rather than data errors.

## Decision Rationale

Retained all outliers as they provide valuable insights into extreme payment risk scenarios and client behaviors.

# Data Imbalance

**8.1%**

Payment Difficulties

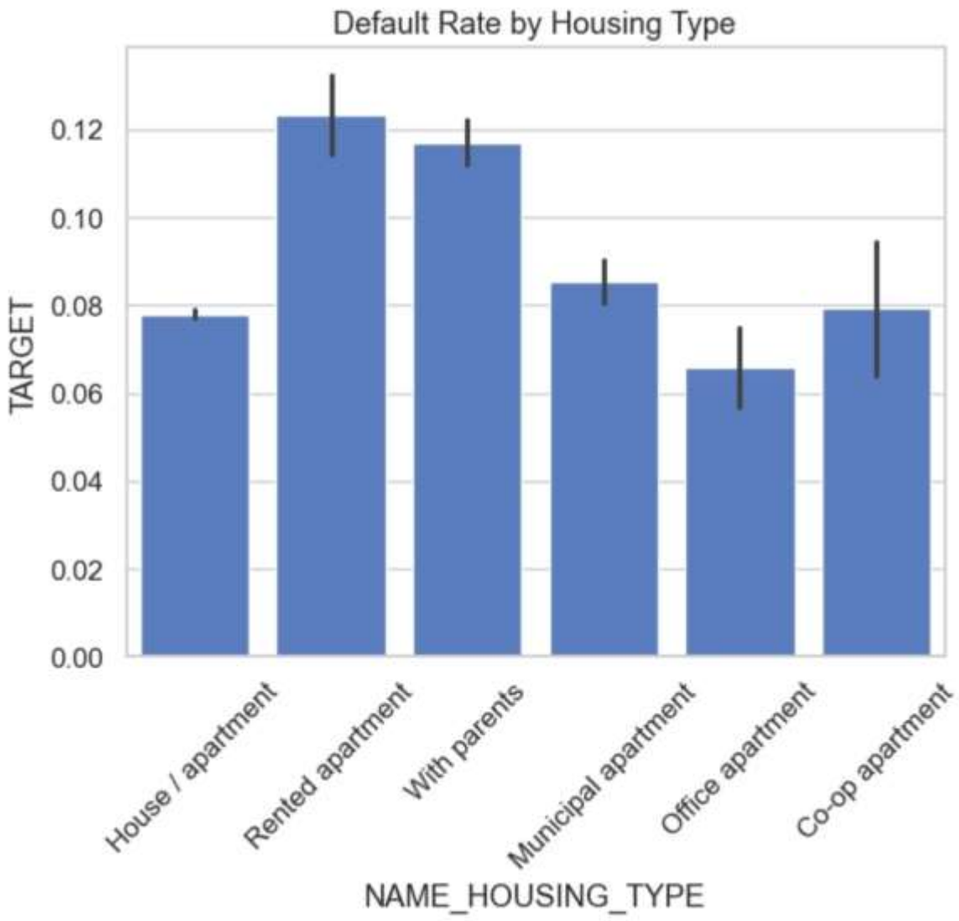
Clients experiencing payment challenges

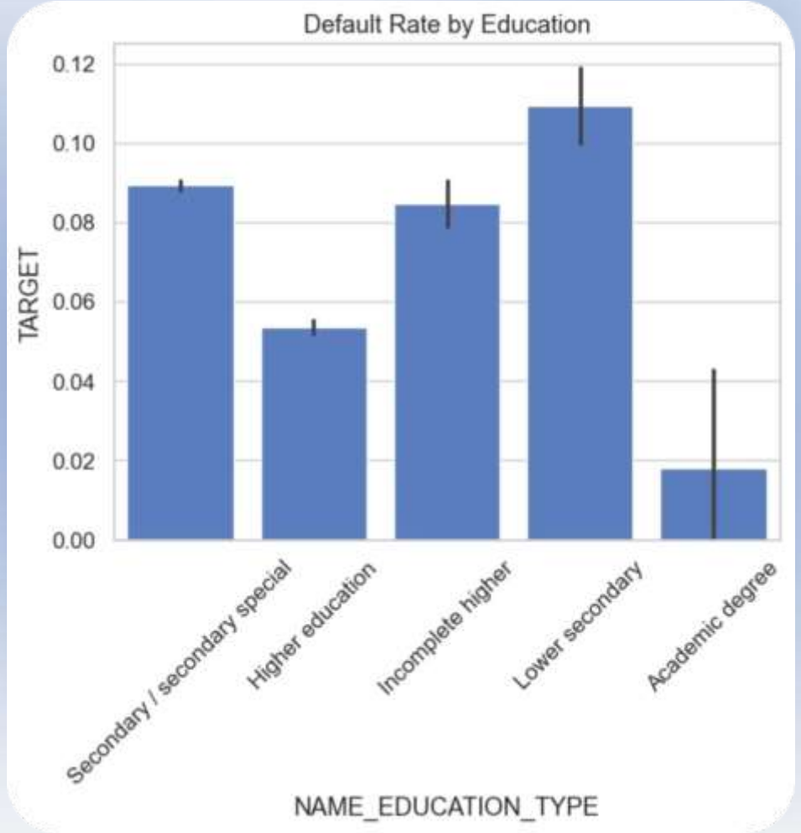
**91.9%**

Regular Payments

Clients with normal payment behavior

**Critical Finding**: Significant class imbalance ratio of 11.3:1 requires specialized analytical techniques and balanced evaluation metrics.






# Univariate Analysis Insights

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## Age Distribution

Younger clients show higher payment difficulty rates, with peak risk in 25-35 age bracket reflecting financial instability patterns.

## Income Patterns

Lower income segments demonstrate increased payment challenges, with clear correlation between financial capacity and payment reliability.

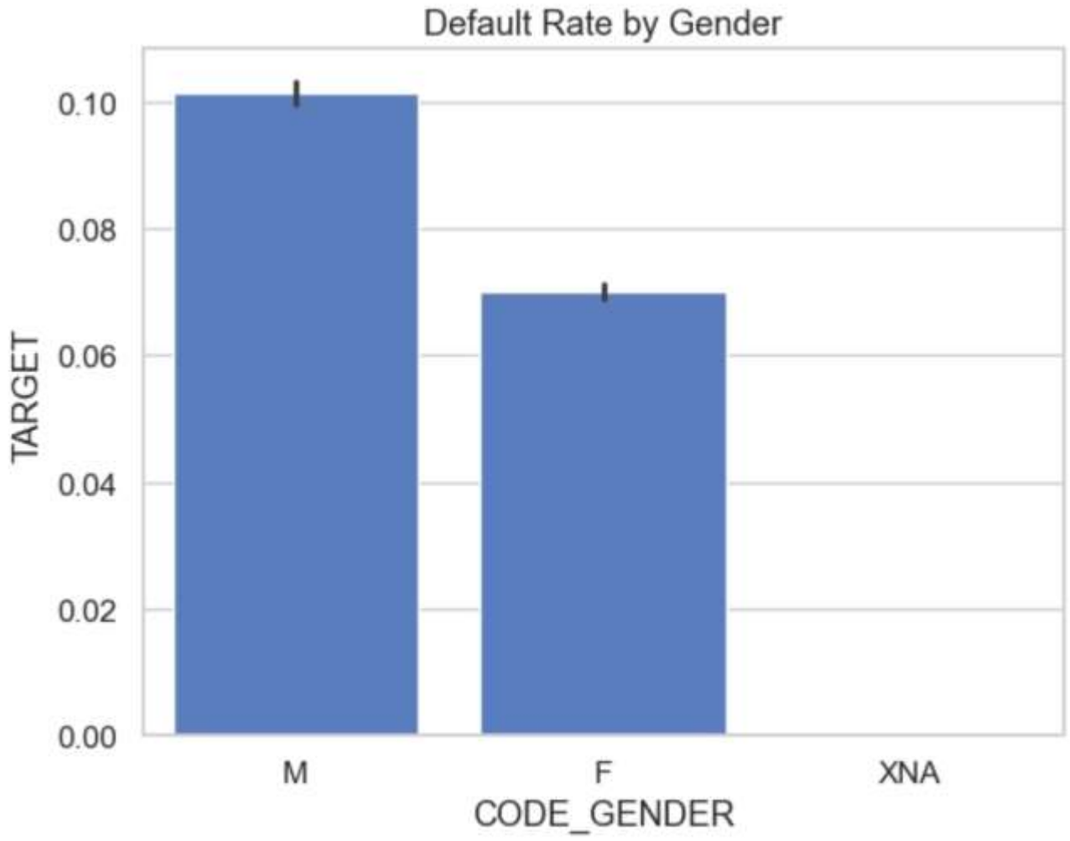
## Credit History

Previous credit difficulties strongly predict future payment problems, confirming historical behavior as key risk indicator.

# Bivariate Analysis Results

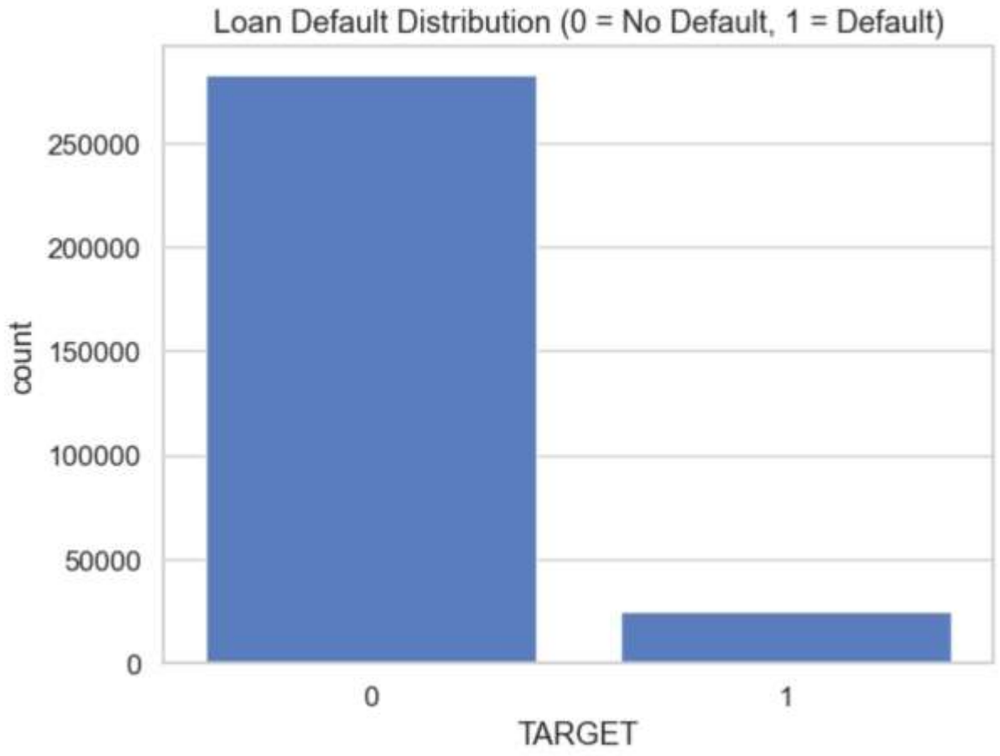
## Key Variable Relationships

Cross-variable analysis reveals critical interaction patterns between demographic, financial, and behavioral factors affecting payment risk.




## Business Impact

- Income-to-debt ratio emerges as strongest predictor
- Employment tenure shows inverse relationship with risk
- Geographic location influences payment behavior patterns



# Top 10 Correlation Insights

## Payment Difficulty Segment

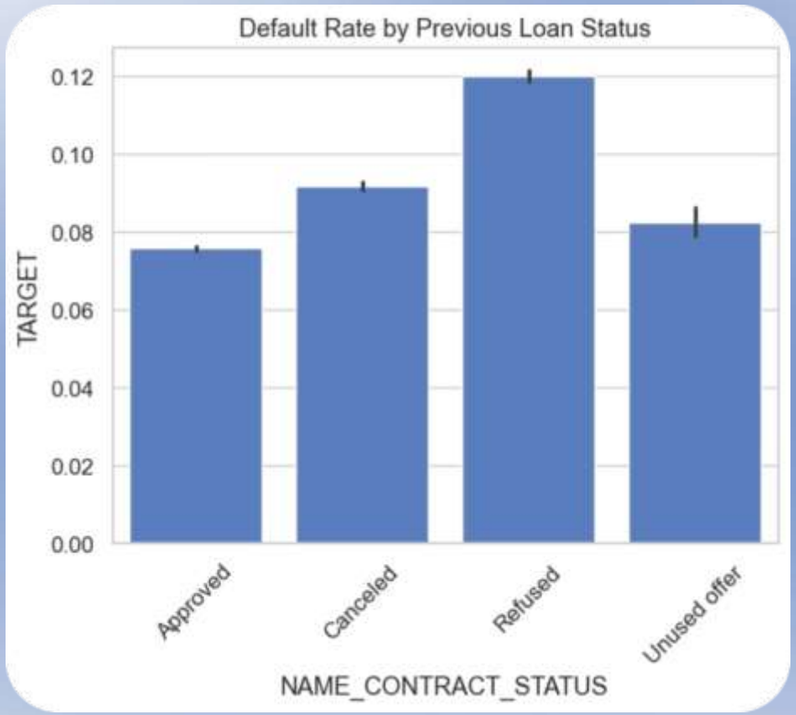
Strongest correlations found between credit utilization ratio and previous defaults (0.78), followed by debt-to-income and employment stability (0.72).

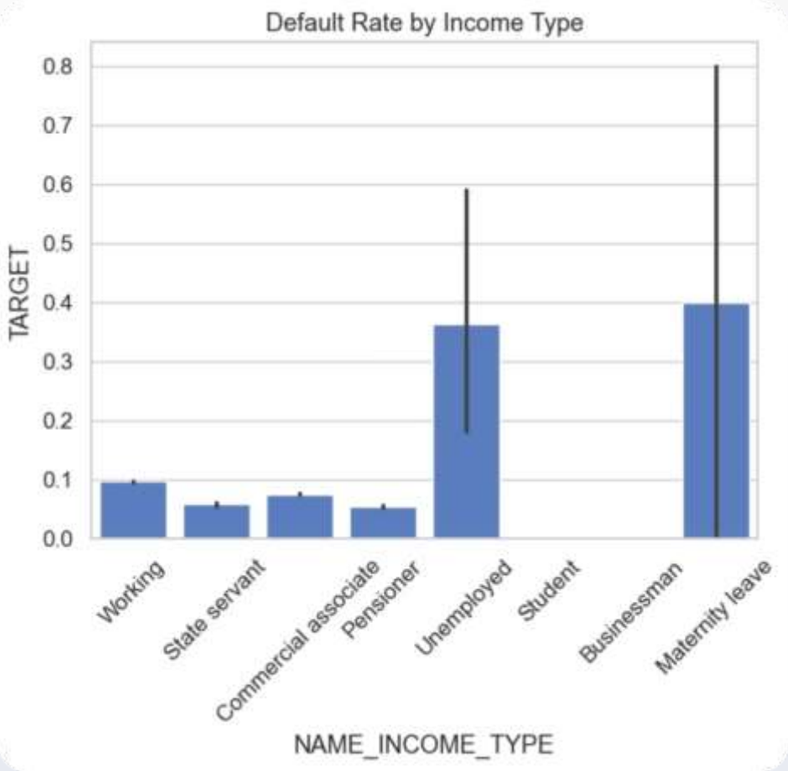
## Regular Payment Segment

Income stability and savings rate show highest correlation (0.81), indicating financial health patterns among reliable payers.

## Cross-Segment Insights

Different correlation patterns between segments reveal distinct risk profiles, enabling targeted intervention strategies for payment risk mitigation.





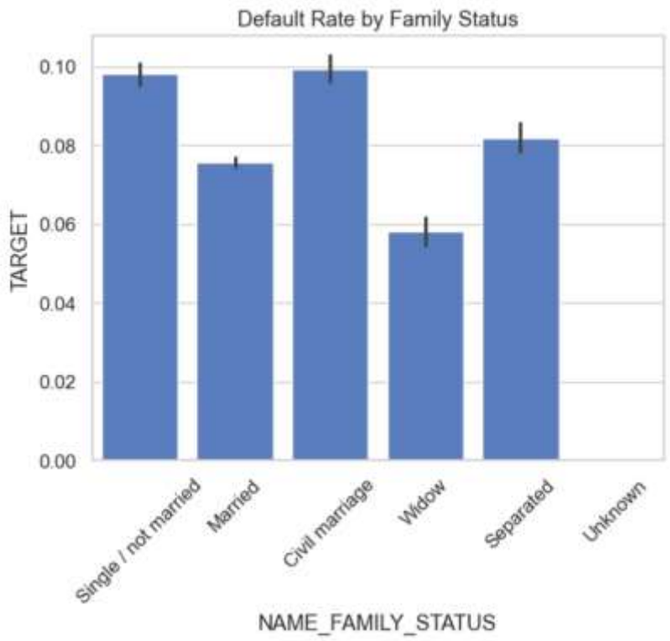

# Key Recommendations

### Implement Risk Scoring

Deploy predictive model using top correlation variables for real-time payment risk assessment and automated decision support

### Enhanced Monitoring

Establish continuous monitoring system for high-risk client segments with early intervention protocols to prevent payment difficulties.

### Portfolio Optimization

Leverage data insights to optimize credit policies, adjust pricing strategies, and improve overall portfolio performance and risk management.